

M-HMS × PowerHub

Integration API Reference

Document version 1.1 — July 2026

PowerHub Production: <https://power.microgenn.com>

M-HMS Production: <https://mhms.microgenn.com>

Overview

PowerHub and M-HMS exchange data over two REST channels using a shared API key agreed between both teams.

Direction	Who calls	Purpose
M-HMS → PowerHub	M-HMS (outbound)	Send ON/OFF commands at guest lifecycle events
PowerHub → M-HMS	PowerHub (outbound)	Pull hotel room master list for import
PowerHub → M-HMS	PowerHub (outbound)	Notify M-HMS when auto-cutoff fires & NEW v1.1

Authentication

Every request in both directions must include the header:

X-API-Key: <shared-api-key>

Missing or invalid key returns 401 Unauthorized. The key is configured once in M-HMS under Configuration → Power Automation, and in PowerHub under Facility → Properties → Edit → MHMS Integration.

1. M-HMS → PowerHub: Send Power Command

M-HMS calls this endpoint on guest lifecycle events (check-in, checkout, etc.).

Request

```
POST https://power.microgenn.com/api/integration/power/commands
X-API-Key: <api-key>
Content-Type: application/json

{
  "roomNumber": "101",
  "action": "ON",
  "controlTypes": ["Light", "AC"],
  "event": "checkin",
  "hotelId": "28fe7680-d1b4-45ae-9949-26f226c3d865",
  "grcNo": "GRC/001/26-27/0000001",
  "guestName": "John Doe",
  "timestamp": "2026-07-22T10:30:00+05:30"
}
```

Field	Type	Required	Description
roomNumber	string	'	Hotel room number e.g. "101"
action	"ON" "OFF"	'	Power action

controlTypes

string[]

Specific types e.g.
["Light","AC"]. Omit = all
controls in the room

event

string

checkin, walkin, checkout,
cleaning, visiting, transfer

hotelId

string

grcNo

string

Guest registration card
number

guestName

string

Guest name (shown in
PowerHub audit trail)

timestamp

string

ISO 8601 with hotel timezone
offset

Response

HTTP 202 Accepted

```
{ "queued": 2, "room": "101", "action": "ON", "event": "checkin" }
```

PowerHub responds 202 immediately. The relay box may take a few seconds to physically execute the command on its next poll cycle.

Events that trigger power commands

M-HMS Event	action	Notes
Check-in / Walk-in	ON	Standard and walk-in check-in
Group check-in / walk-in	ON	Send one request per room in the group
Room transfer	ON + OFF	ON for new room, OFF for old room — two separate requests
Checkout	OFF	Immediate or deferred to settlement
Cleaning	ON	Recommend controlTypes: ["Light"] only
Visiting	ON	Auto-cutoff handles the OFF after visit ends

Error Responses

HTTP Code	Meaning
202	Success — command queued
400	No controls mapped to the room, or invalid control type
401	Missing, empty, or invalid X-API-Key
404	Room number not found in PowerHub
500	Internal server error — log and retry

2. PowerHub !' M-HMS: Pull Room Master List

PowerHub calls this endpoint to import all active rooms. Triggered manually from Facility !' Rooms !' Import from MHMS.

Request

```
GET https://mhms.microgenn.com/api/integration/power/rooms
X-API-Key: <api-key>
```

Response

```
{
  "hotelId": "28fe7680-d1b4-45ae-9949-26f226c3d865",
  "hotelName": "GM Residency",
  "rooms": [
    { "roomNumber": "101", "roomType": "Standard", "block": "Main Block", "floor": "First Floor" },
    { "roomNumber": "106", "roomType": "Deluxe", "block": "Main Block", "floor": "First Floor" }
  ]
}
```

Field	Type	Description
hotelId	string	M-HMS hotel identifier
hotelName	string	Hotel name

```
rooms[].roomNumber
```

string

Room number



rooms[].roomType

string | null

Room type name (e.g. Standard, Deluxe)

rooms[].block

string | null


```
rooms[].floor
```

string | null

Floor name (e.g. First Floor)

Only active rooms should be returned. Deactivated or removed rooms must be excluded.

3. PowerHub !' M-HMS: Auto-Cutoff Status Notification & NEW v1.1

When PowerHub's auto-cutoff engine switches off a room's power (e.g. after a visiting period or cleaning session expires), it immediately notifies M-HMS so the room chart can update its icon and colour in real time.

M-HMS MUST implement this endpoint:

Endpoint M-HMS must implement

```
POST https://mhms.microgenn.com/api/integration/power/status
```

```
X-API-Key: <api-key>
```

```
Content-Type: application/json
```

```
{
  "roomNumber": "101",
  "action": "OFF",
  "event": "auto-cutoff",
  "grcNo": "GRC/001/26-27/0000001",
  "guestName": "John Doe",
  "timestamp": "2026-07-22T12:30:00.000Z"
}
```

Field	Type	Description
roomNumber	string	Room number that was switched off
action	"OFF"	Always "OFF" for auto-cutoff notifications
event	string	Always "auto-cutoff" for this notification type
grcNo	string null	GRC number of the original check-in/visit, if available
guestName	string null	Guest name of the original command, if available
timestamp	string	ISO 8601 UTC timestamp when cutoff was executed

Expected Response from M-HMS

```
HTTP 200 OK (any 2xx is accepted)
```

PowerHub fires this call fire-and-forget — it does not block the cutoff on a response. If M-HMS is unreachable, PowerHub logs a warning and continues. M-HMS must implement this endpoint so the room chart reflects the correct OFF state without waiting for a manual refresh.

Suggested M-HMS behaviour on receiving this notification

- Update the room power status indicator to OFF / Standby.
- Change the room card colour from green (live) to grey/red (standby).
- Update the icon to reflect no active power load.
- Log the auto-cutoff event in the M-HMS audit trail for the GRC.

One-Time Setup

In M-HMS

1. Log in !' Configuration !' Power Automation.
2. Fill in PowerHub Base URL: <https://power.microgenn.com>

3. Fill in API Key: shared key agreed with PowerHub team.
4. Enable Power Automation and click Save.
5. Use Test Connection (enter a room number, click Test Power ON) to verify.

In PowerHub

1. Log in !' Facility !' Properties !' Edit the property.
2. Scroll to the MHMS Integration section.
3. Fill in MHMS Server URL: <https://mhms.microgenn.com>
4. Fill in MHMS API Key: same shared key.
5. Click Save Property.
6. Go to Facility !' Rooms !' Import from MHMS to pull the room master list.

Staging / Test Environment

For testing before going live, use the staging M-HMS server. Authentication and all endpoints are identical to production.

Environment	URL
M-HMS Staging	https://staghms.microgenn.com
PowerHub Production	https://power.microgenn.com

Companion Bridge (On-Premises Relay Boxes)

If the hotel's relay boxes (ESP32 firmware) are on a local LAN and cannot reach power.microgenn.com directly, install the PowerHub Companion Bridge on any always-on Windows PC at the hotel.

Download: Log in to PowerHub !' Integration !' Power Automation !' click Download Bridge.

The bridge accepts plain HTTP from the relay boxes and forwards it to PowerHub over HTTPS. No firewall changes are needed on the PowerHub side.